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Skin Tone Analysis Through Skin Tone Map Generation With Optical Approach and Deep Learning

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Keywords: deep learning | hemoglobin | ITA | melanin | skin tone

ABSTRACT

Background: Skin tone assessment is critical in both cosmetic and medical fields, yet traditional methods like the individual typology angle (ITA) have limitations, such as sensitivity to illuminants and insensitivity to skin redness.

Methods: This study introduces an automated image-based method for skin tone mapping by applying optical approaches and deep learning. The method generates skin tone maps by leveraging the illuminant spectrum, segments the skin region from face images, and identifies the corresponding skin tone on the map. The method was evaluated by generating skin tone maps under three standard illuminants (D45, D65, and D85) and comparing the results with those obtained using ITA on skin tone simulation images.

Results: The results showed that skin tone maps generated under the same lighting conditions as the image acquisition (D65) provided the highest accuracy, with a color difference of around 6, which is more than twice as small as those observed under other illuminants. The mapping positions also demonstrated a clear correlation with pigment levels. Compared to ITA, the proposed approach was particularly effective in distinguishing skin tones related to redness.

Conclusion: Despite the need to measure the illuminant spectrum and for further physiological validation, the proposed approach shows potential for enhancing skin tone assessment. Its ability to mitigate the effects of illuminants and distinguish between the two dominant pigments offers promising applications in both cosmetic and medical diagnostics.

1 | Introduction

Skin tone is a prominent feature of human appearance that varies significantly across populations and individuals. This variation is primarily attributed to the absorption spectra of major skin pigments: melanin, which imparts darker and yellowish tones [1], and hemoglobin, which contributes to redness [2]. Accurate skin tone assessment is vital not only for personalized cosmetic recommendations [3] and personal color selection [4] but also for addressing biases in clinical measurements, such as pulse oximetry, where variations in skin tone can affect accuracy [5].

The individual typology angle (ITA) is a widely used method for skin tone measurement, calculated using the CIE Lab* color space. ITA determines the angle between L* (lightness) and b* (yellow-blue), classifying skin tones into categories ranging from very light to dark [6]. Despite its utility, ITA is limited by its reliance on a single axis, making it less effective at distinguishing

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between the influences of different pigments, such as melanin and hemoglobin. The Fitzpatrick scale is another method that categorizes skin types based on their burning and tanning response to ultraviolet radiation rather than their actual color [7]. Visual methods, such as matching skin tones to printed color tiles [8], are also used but can lack objectivity and reliability due to color changes affected by lighting conditions [9]. Additionally, while tools like colorimetry and spectrophotometry offer precise measurements, they are expensive and less practical compared to camera-based methods.

In this study, we propose an automated image-based skin tone mapping method for skin tone analysis, utilizing an optical approach combined with deep learning. This method involves generating a skin tone map based on the illuminant spectrum during image acquisition, segmenting the skin region in the captured image, and identifying the corresponding skin tone on the map. The effectiveness of the method is evaluated by comparing skin tone maps generated under different illuminants for the same dataset and by comparing the results to those obtained using ITA. This research aims to establish a fundamental principle for skin tone classification under various lighting conditions, offering improved performance over existing methods.

2 | Methods

The proposed approach and the ITA method, used here for comparison, are illustrated in Figure 1. Both methods begin with the detection of the skin region using a skin segmentation model. In the proposed approach (Figure 1A), a skin tone map is generated based on the spectral characteristics of the illuminant, skin reflectance, and the standard observer's functions. In contrast, the ITA method (Figure 1B) involves calculating the angle between the L* (lightness) and b* (yellow-blue) axes in the Lab* color space. Detailed procedures and a comparative analysis of both methods are provided below.

2.1 | Skin Tone Mapping

The entire process consists of three main components: the results from the skin segmentation model and the skin tone map generation parts are integrated by the skin tone mapping module (Figure 1A). The skin tone map is based on the spectral power distribution (SPD) of the illuminant present in the image acquisition environment. For generalization purposes, this study employs the standard illuminant D series (daylight) to create skin tone maps.

2.1.1 | Skin Segmentation Model

Excluding nonskin regions from face images is essential for improving the accuracy of average skin tone measurements. However, manually extracting skin regions from a large number of images is impractical. To address this challenge, we implemented an automated skin segmentation model.

The model is based on the ResUNet++ architecture, which has been demonstrated to outperform the original UNet structure [10]. The network was optimized by reducing the number of parameters from 16.23 to 4.06 million, decreasing the computational cost (GFLOPS) from 294.84 to 74.12 GMac, and achieving an inference time of 19.27 ms on an NVIDIA GeForce RTX 3090. The network was trained on a dataset of 2329 images, each with a resolution of 640 × 480 pixels, to differentiate between skin and nonskin regions (such as eyebrows, hair, nostrils, lips, and background) by generating binary masks. The model achieved a complete intersection over union (CIoU) [11] of 0.963, with sensitivity, specificity, and F1-scores [12] of 0.983, 0.980, and 0.980, respectively.

Figure 2A presents the results from the trained model. The skin segmentation mask is generated from the input image and overlaid onto the original image to isolate the skin region. The results indicate that the model successfully excluded most nonskin regions.

2.1.2 | Spectral Skin Tone Generation

The process by which illuminated light is reflected from the skin and perceived by the eyes can be spectrally modeled as follows [13]. When light hits the skin, a small portion (approximately 4% to 7% [14]) is specularly reflected, whereas the majority is refracted into the skin layers. The refracted light undergoes absorption and scattering within the skin layers, which can be mathematically described as follows [15]:

$$\mu_a(\lambda) = \Sigma_i V_i \epsilon_i(\lambda) C_i \tag{1}$$

$$\mu_s'(\lambda) = a\left(f_{Ray}\left(\frac{\lambda}{\lambda_0}\right)^{-4} + (1 - f_{Ray})\left(\frac{\lambda}{\lambda_0}\right)^{-b_{Mie}} \right) \tag{2}$$

$\mu_a(\lambda)$ represents the absorption coefficient, V_i denotes the volume fraction, ϵ_i is the extinction coefficient, and C_i is the concentration of chromophores. In this study, the chromophores are assumed to be the primary pigments—melanin and hemoglobin—within a wavelength range (λ) of 450 to 750 nm. The term $\mu_s'(\lambda)$ represents the reduced scattering coefficient, where a is the scattering amplitude, b is the scattering power, and f_{Ray} is the fraction of Rayleigh scattering. This equation accounts for both Rayleigh and Mie scattering.

Skin reflectance can be determined through analytic methods, such as diffuse reflectance spectroscopy [15] or the Kubelka–Munk theory [16], or through stochastic approaches like Monte Carlo simulations [17]. In this study, the skin was simplified into epidermis and dermis layers using the Kubelka–Munk theory, and the calculated reflectance data, validated by previous research [18], were employed. The volume fractions for melanin range from 1.3% to 43%, and for hemoglobin from 2% to 7% [19, 20].

Multiplying the SPD of the illuminants by the skin reflectance across all wavelengths yields the light reflected from the skin. When this result is multiplied by the stimulus functions of the CIE 1931 2 Degree Standard Observer [21], it produces color

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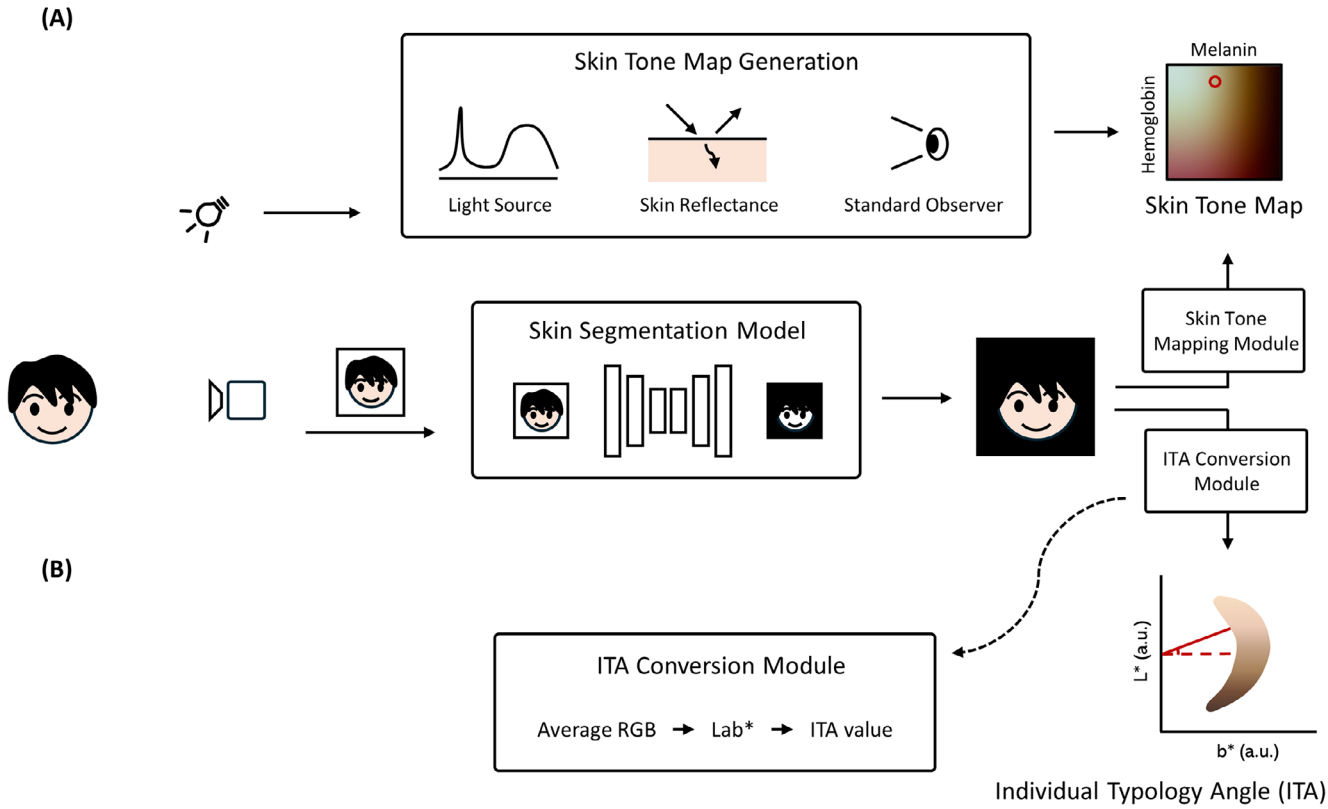

FIGURE 1 | Schematic diagram of the image-based skin tone mapping methods. (A) The proposed approach consists of two components: skin tone map generation and skin segmentation. The skin tone map, which is perceived by the eyes, is modeled by combining the spectral power distribution of the illuminant, skin reflectance, and the stimulus functions of the standard observer. The skin reflectance is primarily influenced by melanin and hemoglobin, which form the axes of the generated map. The skin segmentation model detects the skin region in a facial image, and the skin tone mapping module identifies the average skin color and its closest match on the generated skin tone map. (B) The ITA method calculates the angle formed by L* and b* in the Lab* color space, based on the average color of the segmented skin region. ITA, individual typology angle.

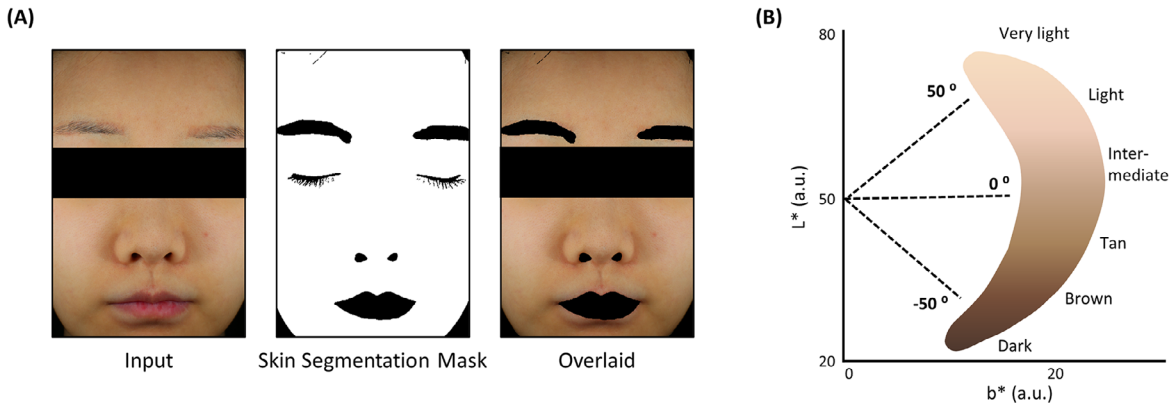

FIGURE 2 | Illustration of skin segmentation and the ITA concept. (A) The input to the skin segmentation model is a face image, which produces a binary mask as the output. This mask is then overlaid on the original image, with the skin regions displayed in white and the nonskin regions, including the background, shown in black. (B) The ITA diagram illustrates how skin tones are categorized based on the angle between L* and b* in the Lab* color space, with L* = 50 serving as the reference point for representing skin tone. ITA, individual typology angle.

values in the XYZ color space. These values are then transformed into the RGB color space using a transformation matrix, and gamma correction is applied to convert them into the sRGB color space, ultimately providing the color perceived by the human eye.

2.1.3 | Skin Tone Mapping

For skin tone mapping, the average RGB value of the skin region is first calculated and then converted into the Lab* color

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space. The Lab* color space is more suitable for identifying how different colors appear to the human observer because it better reflects actual visual perception compared to the general RGB models [22]. The Euclidean distance between two colors is used to quantify the color difference, expressed by the following equation:

$$\Delta E^* = \sqrt{(L_2^* - L_1^*)^2 + (a_2^* - a_1^*)^2 + (b_2^* - b_1^*)^2} \quad (3)$$

L_i^*, a_i^*, b_i^* are the L^*, a^*, b^* values of the i-th colors, and ΔE^* represents the color difference between the two colors.

In the skin tone mapping module, the RGB values of all points on the skin tone maps are converted into the Lab* color space. The color difference between each point and the average Lab* value of the skin region is then calculated. The location and color with the smallest color difference are identified, representing the closest match to the actual skin tone of the skin region. However, to ensure accuracy, it is important to verify the color difference value, as an incorrect illuminant could cause the overall color to differ from the actual skin tone, and potentially leading to inaccuracies.

2.2 | ITA

ITA is a metric used to quantify skin tone based on lightness and yellowness, calculated using the following equation:

$$ITA = \arctan\left(\frac{L^* - 50}{b^*}\right) \times \frac{180^\circ}{\pi} \quad (4)$$

L^* represents lightness, ranging from black to white, and b^* represents the blue to yellow axis. The angle is calculated using a reference point of $L = 50$, and the result is converted from radians to degrees, a more intuitive unit for interpretation.

Based on the ITA angle, skin tone is generally classified into the following categories (Figure 2B): very light (above 55°), light (41° to 55°), intermediate (28° to 41°), tan (10° to 28°), brown (−30° to 10°), and dark (below −30°) [23].

2.3 | Experiments

We obtained face images with a resolution of 3648 × 5472 pixels, captured using the VISIA system (Canfield Scientific Inc.) from the Global Medical Research Center. These images were cropped and resized to 915 × 1250 pixels for use in the study. Data were acquired under standard illuminant D65 conditions from 25 subjects, following national regulations and institutional research ethics policies. All participants provided informed consent, and the study received approval from the institutional review board of the authors' institution.

Modifying a person's actual skin pigment levels is practically challenging. Therefore, to prepare the dataset used for validation, we generated skin tone simulation images based on our previous research [24–27]. Specifically, we developed a framework using a deep learning model to analyze the distribution of melanin and hemoglobin pigments in skin images, followed by reconstructing this pigment distribution in the skin image using an analytical approach. The performance of this model was validated in previous studies by comparing its results with those obtained from professional diagnostic systems. Simulated images were created by modifying the skin tone, and adding a constant to the pigment levels in the intermediate steps of the framework. The pigment variations ranged from −0.6 to +0.6 (with an interval of 0.1), expressed in arbitrary units.

Using these acquired face images and skin tone simulation images, we conducted two evaluations. First, skin tone maps generated from three standard illuminants (D45–warm, D65–standard, and D85–cool light) were applied to the same face images captured under D65 lighting conditions. The objective was to verify whether the skin tone map generated under the corresponding illuminant condition exhibited superior performance. The relationship between the coordinate changes (x- and y-axis) of the skin tone mapping positions and the pigment changes (for melanin and hemoglobin, respectively) was analyzed. Additionally, the color difference between the mapped color and the segmented skin color was quantified as an accuracy metric.

Next, the proposed method was compared with ITA to identify the characteristics of each approach. The skin tone map was directly converted to an ITA map, and their trends were qualitatively compared. We examined the skin tone mapping method's ability to distinguish between subjects with similar ITA angles. We also discussed changes in ITA values and mapping positions in relation to varying pigment levels.

Further details of the training and computing conditions are as follows. For training the skin segmentation model, we used the Adam optimizer (with a weight decay of l2 = 0.001) and a learning rate scheduler (cosine annealing, T = 100), starting with a learning rate of 0.0001, which decreased to a minimum of 0.000001. We employed a batch size of 12 and cross-entropy loss, with data augmentation performed using the Albumentations library [28], incorporating techniques such as horizontal flip, shift-scale rotation, random crop, hue saturation, random brightness contrast, unsharp mask, and image compression. Spectral analysis was conducted using the Colour-Science library [29], and its SPD data were utilized. The code was implemented in Python using the PyTorch platform and executed in PyCharm software on an RTX 3090 GPU.

3 | Results

Figure 3 presents the results of applying skin tone maps generated for three standard illuminants (D45, D65, and D85) to the simulation images. Figure 3A,B shows the generated skin tone maps and the simulation images with modified melanin and hemoglobin levels (−0.6 and +0.6), respectively. Figure 3C illustrates the changes in skin tone mapping positions due to pigment level variations, while Table 1 summarizes the average mapping position changes across all pigment levels. Furthermore, Table 2 evaluates the mapping accuracy by showing the average color difference between the mapped color and the segmented skin color across seven hemoglobin and melanin levels.

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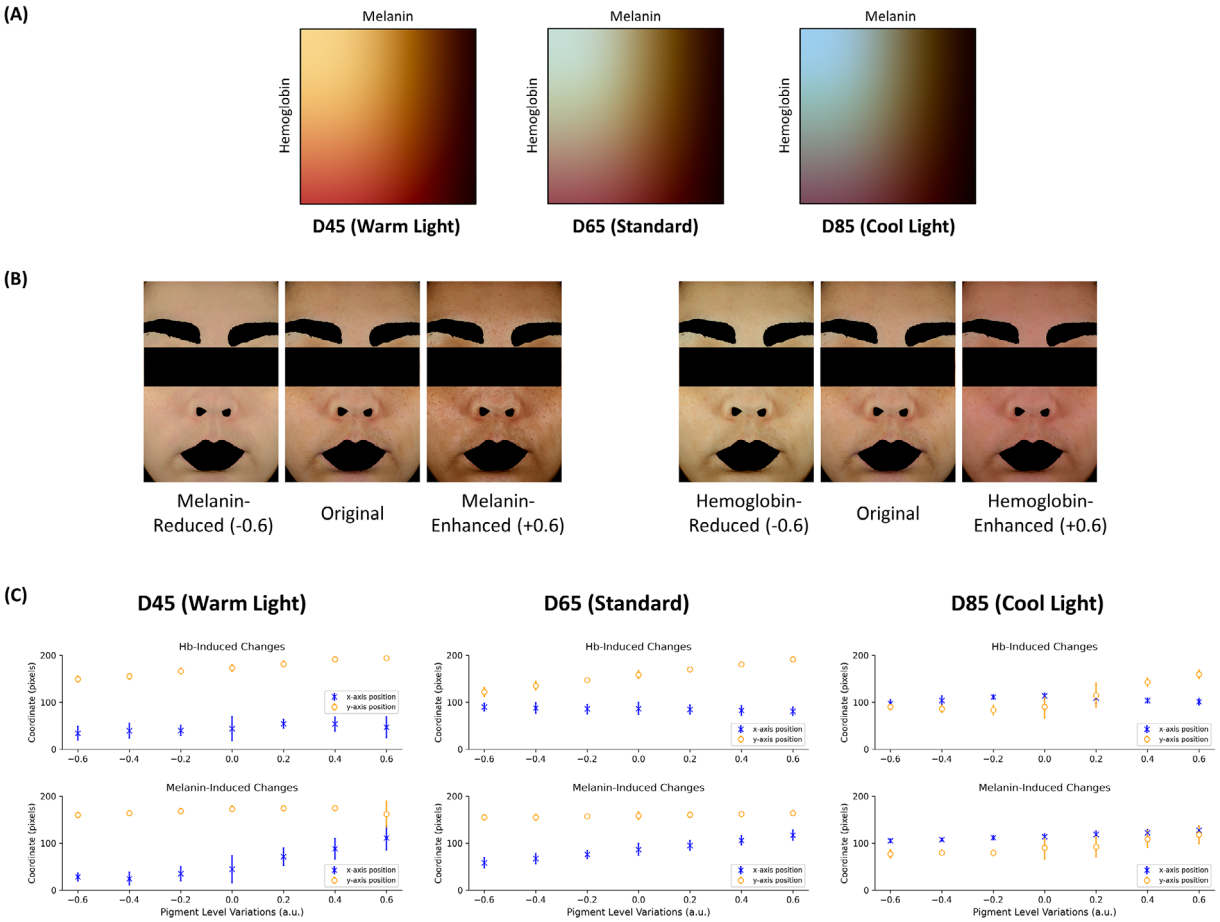

FIGURE 3 | Application of skin tone maps to simulation images. (A) Skin tone maps generated under three standard illuminants: D45 (warm light), D65 (standard), and D85 (cool light). The *x*- and *y*-axes correspond to melanin and hemoglobin, respectively. (B) Pigment simulation images showing reduced (−0.6) and enhanced (+0.6) (left) melanin and (right) hemoglobin levels. (C) Coordinate changes (blue = *x*-axis, yellow = *y*-axis) in the skin tone map due to pigment level variations (−0.6 to +0.6) for the simulation images. ITA, individual typology angle.

TABLE 1 | Average coordinate changes in skin tone mapping positions for three illuminants, corresponding to variations in hemoglobin and melanin levels.

	D45	D65	D85
Hb	x: 1.20 ± 17.30	x: −1.47 ± 6.75	x: 0.60 ± 6.90
	y: 8.97 ± 4.61	y: 11.55 ± 6.78	y: 11.53 ± 17.15
Mel	x: 12.40 ± 15.60	x: 9.74 ± 7.86	x: 3.80 ± 5.80
	y: 1.97 ± 5.06	y: 1.46 ± 5.88	y: 7.15 ± 19.18

TABLE 2 | Average color difference between the mapped color and the segmented skin color for variations in hemoglobin and melanin levels.

	D45	D65	D85
Hb	13.58 ± 4.24	6.20 ± 1.35	16.96 ± 2.47
Mel	14.24 ± 4.36	6.03 ± 1.25	16.27 ± 2.11

Figure 4 compares the proposed method with the ITA approach. Figure 4A shows the skin tone map colors converted to ITA degrees, demonstrating a range of pseudo colors from dark to light skin tones. Figure 4B compares skin tone mapping positions for subjects with similar ITA angles. Figure 4C illustrates the changes in ITA and skin tone mapping positions in response to pigment level variations. The ITA angles range from approximately 5 to 40, with standard deviations averaging 4.31 for both hemoglobin and melanin distributions. In contrast, the skin tone mapping positions vary within a range of about 100 pixels, with standard deviations averaging 11.55 for hemoglobin (*y*-axis) and 9.74 for melanin (*x*-axis).

4 | Discussion

To evaluate the performance of skin tone maps generated under the same lighting conditions as those used during image acquisition, we applied skin tone maps for three standard illuminants (D45, D65, and D85) to images captured under D65 lighting conditions, as shown in Figure 3. For simplicity, we will refer to "standard illuminant D65," "D45," and "D85" simply as "D65," "D45," and "D85." These three illuminants were selected because D65 is typically used as the standard for color measurement, while D45 and D85 represent warm and cool light conditions, respectively, accounting for variations in color temperature. As illustrated in Figure 3A, skin tone maps for different color

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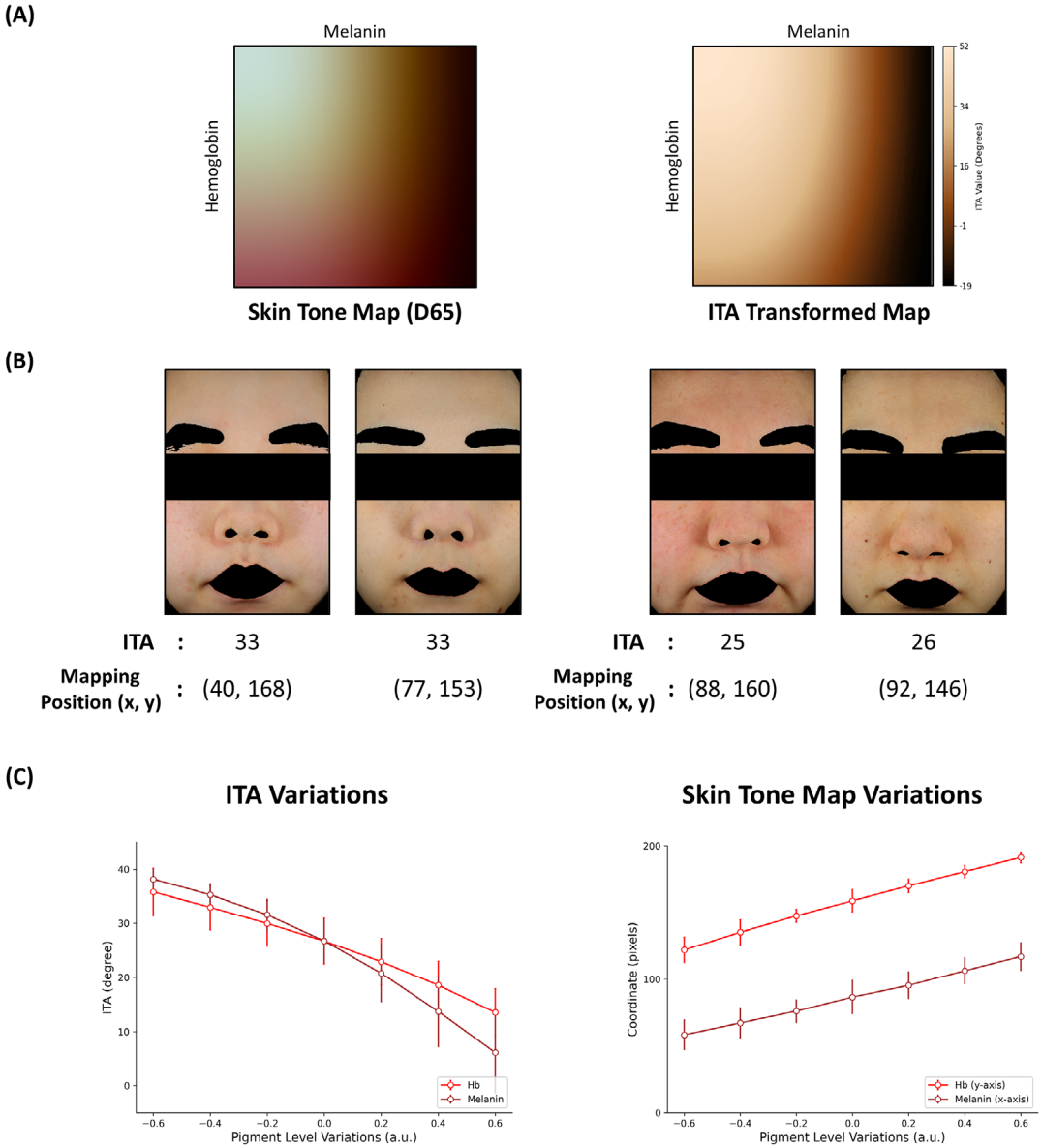

FIGURE 4 | Comparison of the proposed method with ITA. (A) (left) The skin tone map and the (right) ITA transformed map, where the skin tone map's colors have been converted into ITA degrees to illustrate a range of pseudo colors from dark to light skin tones. (B) Skin tone mapping positions for subjects with similar ITA angles (left: 33, right: 26). (C) Comparison of ITA (left) and skin tone mapping positions (right) in response to pigment level variations. ITA, individual typology angle.

temperature conditions are shown based on variations in melanin and hemoglobin levels. Figure 3B demonstrates the skin tone simulation images: (left) variations in melanin levels affect skin darkness, and (right) variations in hemoglobin levels influence skin redness. In the hemoglobin-reduced image, the hemoglobin level drops nearly to zero, making melanin the predominant influence, resulting in a yellowish appearance. Pigment levels were restricted to the range of −0.6 to +0.6 to avoid unnatural images that could result from excessive or insufficient pigment saturation.

The application of the three color maps to the simulation images was evaluated by analyzing the coordinate changes in skin tone mapping positions and the color difference between the mapped color and the segmented skin tone. In the skin tone map, variations in melanin levels generally shifted the *x*-axis, while hemoglobin levels affected the *y*-axis (Figure 3C). Under D45, the standard deviation of the *y*-axis increases as melanin levels rise. Under D85, variations in both axes are similar as melanin levels change, with the standard deviation of the *y*-axis widely spread for both hemoglobin and melanin changes. In contrast, D65 shows a clearer correlation with the pigment levels, with the standard deviation remaining consistent across different variation levels. This trend is also numerically supported by the average changes across all pigment levels, as shown in Table 1, indicating that the skin tone map effectively reflects pigment variations in the face images. Additionally, Table 2 provides a metric for assessing mapping accuracy by showing the average color difference between the mapped color and the segmented skin color across all pigment levels. The color difference under D65 is approximately 6, which is more than twice as small as those observed under D45 and D85 for both hemoglobin

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and melanin level variations, indicating that the skin tone map generated under lighting conditions matching the image capture environment provides the highest accuracy.

The comparison between the skin tone map and ITA is presented in Figure 4. Figure 4A shows the ITA transformed map, where the colors of the skin tone map have been converted to ITA angles and assigned pseudo colors. Compared to the skin tone map, the ITA transformed map demonstrates relative insensitivity to changes in hemoglobin levels. This can be attributed to ITA's primary focus on lightness and yellowness, which are more influenced by melanin-induced darkness. Figure 4B shows the mapping positions for subjects with similar ITA angles. Unlike ITA, which relies on a single axis (angle), the skin tone map uses two axes—melanin and hemoglobin—allowing it to distinguish between different skin tones, particularly in terms of redness. Figure 4C displays graphs of ITA and mapping positions as a function of pigment level variations. The average distribution range of ITA angles is 9.8° wider for melanin (32.07) than for hemoglobin (22.27), suggesting that ITA is more sensitive to changes in melanin. Additionally, as pigment levels increase, the standard deviation also increases, implying that ITA may be less stable for tracking pigment changes. The tendency for ITA to increase as pigment levels decrease is consistent for both pigments, making it difficult to determine which pigment is influencing the ITA value. In contrast, the skin tone map shows a clearer correlation, with mapping position coordinates increasing proportionally with the corresponding pigment levels while maintaining a consistent standard deviation. This proportionality in coordinate changes suggests that the skin tone map can more effectively distinguish the effects of the two pigments compared to ITA, offering potential for diagnostic applications where stability and accuracy across different pigment levels are crucial.

However, the proposed approach has some limitations. First, if the measurement environment is unknown, it may be necessary to use a spectrometer to measure the illuminant. In this study, we used a professional device with a standard illuminant, allowing us to use known spectrum data. Additionally, for universal application in the future, further validation is needed not only for standard illuminants but also for combinational illuminants such as fluorescent lights and sunlight. Since it is practically difficult to replace the light sources installed in high-end professional equipment, we conducted evaluations using images captured under D65 only. Moreover, we assumed a linear relationship between pigment levels and actual pigment quantities when generating the simulation images, but we did not physiologically validate this assumption. We only performed indirect validation by examining the correlation with diagnostic results from professional equipment. If the proposed approach is to be expanded into the diagnostic field, further validation will be necessary.

5 | Conclusions

In this study, we introduced an automated image-based method for skin tone mapping. This method involves generating a skin tone map based on the illuminant spectrum, segmenting the skin region in the acquired face image, and identifying the corresponding skin tone on the map. Our evaluation confirmed that the skin tone map generated under the same lighting conditions as those used during image acquisition (standard illuminant D65) provides the highest accuracy. Additionally, this approach effectively addresses some limitations of the traditional ITA method, particularly its insensitivity to skin redness. However, the proposed method has certain limitations, including the need to know the illuminant spectrum in advance and the requirement for further physiological validation. Despite these challenges, the method offers advantages that reflecting the effects of melanin and hemoglobin on skin tone in controlled environments where the illuminant is known. With further development, this method has the potential to become a valuable tool in both cosmetic and medical applications, enabling more accurate and individualized skin tone assessments.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data associated with this article are not publicly available due to privacy and confidentiality considerations. Data can be requested for research purposes through official collaborations with our company, directed to the corresponding author, Sangwook Yoo. Prior consent from the subjects and the necessary approvals from the author's institutional review committees will be required.

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